

Course Code: BCC 108
Course Title: Business Statistics
Marks: 100
Credit: 3 Hours
(Written: 75, Internal Evaluation: 20, and Viva: 05)

Course Objectives:

The course provides insight into the statistical processes of data classification, description, presentation, and association.

Learning Objectives:

1. To learn the basic concept of statistics.
2. To understand the nature of data and the presentation of data in different forms.
3. To measure central tendency and dispersion of data.
4. To learn the probability calculation and its application.
5. To learn how to compute correlation and regression between variables.
6. To learn how to do forecasting and measures of index number.
7. To learn tools for statistical process control

Course Contents:

1. **Introduction:** Meaning of statistics; Business statistics; Types of statistics; Types of variables; levels of measurement; Uses of statistics in business; Limitation of statistics.
2. **Describing Data (Tabular and Graphical Presentation):** Constructing of frequency distribution; Bar chart; Pie chart; Frequency histogram; Frequency curve; Frequency polygon; ogive curve.
3. **Describing Data (Numerical Measures):**
 - a. Measures of Central Tendency: Definition, mean, median, mode and weighted mean.
 - b. Measures of Dispersion: Definition, types, range, variance, standard deviation, coefficient of variation.
4. **A Survey of Probability Concept:** Meaning; Different of Approaches; Probability Rules, Bayes' Theorem.
5. **Probability Distribution:** Definition, Random variable, Mean, Variance and Standard Deviation of a Probability distribution; Definition, characteristics and examples of Binomial Probability Distribution; Poisson Probability distribution; Normal Probability distribution.
6. **Correlation and Linear Regression:** Introduction; Correlation Analysis; The Coefficient of Correlation; The Coefficient of Determination; Significance of the Study of Correlation and Causation; Types of Correlation; Scatter Diagram Method; Karl Pearson's Coefficient of Correlation; Rank Correlation Co-efficient; Regression analysis: Least square principle; Drawing the Line of Regression; Problems regarding regression analysis.
7. **Time Series and Forecasting:** Introduction; Components of Time Series; Measures of Trend; Graphic Method; Method of Semi Averages; The moving average Method; Least Squares Method; Seasonal Variation; Deseasonalizing Data; Using deseasonalized data to forecast.

8. **Index Number:** Introduction; Simple Index Numbers; Why Convert data to Indexes? Constructions of Index Numbers; Weighted Indexes; Unweighted Indexes; Value Index; Special Purpose Indexes; Consumer Price Index; Shifting the Base. 9. **Statistical Process Control:** Types of Control Charts; Mean Charts; Range Chart; C Chart; p Chart; Benefits and Limitations of Statistical Quality Control.

Basic Texts:

1. Lind, Marchal & Walther; Statistical Techniques in Business and Economics

Reference Books:

1. Lind, Mason, & Marchal; Basic Statistics for Business and Economics.
2. Richard Levin & David S. Rubin, Statistics for Management.
3. M.K Roy and J. C Paul, Business Statistics.